

TANMAY KUMAR SAHU

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Professional Summary

GATE-qualified IT undergraduate with hands-on experience building **LLM-powered applications**, end-to-end ML pipelines, and **RAG architectures** using LangChain, Hugging Face, and vector databases. Proficient in **Python**, **REST APIs (FastAPI)**, and deploying AI solutions as production-ready systems. Strong foundation in transformer architectures, embeddings, and prompt engineering. Demonstrated ability to translate complex problems into structured, scalable AI solutions.

Education

Guru Ghasidas Vishwavidyalaya, Bilaspur

Bachelor of Technology in Information Technology (CGPA: 7.81)

Dec 2022 – May 2026

Bilaspur, Chhattisgarh

J.L.M. Gayatri Vidya Peeth, Rajnandgaon

CBSE Class 12th – PCM (88.6%)

Jun 2021

Rajnandgaon, Chhattisgarh

Technical Skills

- **Languages & Technologies:** Python, SQL, C, FastAPI, Streamlit, Git, Jupyter Notebook
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Librosa, LangChain, Hugging Face, Gemini API, Pandas, NumPy, Matplotlib, Seaborn, SciPy, FAISS, ChromaDB
- **Competencies:** RAG, Vector Databases, Prompt Engineering, Data Analysis, Machine Learning, Foundation Models, Multi-agent Systems

Projects

Enterprise Document Intelligence Chatbot (RAG)

LangChain, ChromaDB, Gemini API, Streamlit, Python

Mar 2026 – Apr 2026

GitHub

- Built a **RAG pipeline** using **LangChain** + **ChromaDB** vector store for semantic search over enterprise PDF knowledge bases.
- Integrated **Gemini API** with **prompt engineering** templates for grounded, citation-backed LLM responses.
- Deployed as a **Streamlit** app with document upload, chunk retrieval, and a conversational Q&A interface.

Bridging Traditional Rhythms and Digital DAWs with Machine Learning

Audio ML, PyTorch, Librosa

Jan 2026 – Apr 2026

GitHub

- Built a **CNN + BiLSTM** pipeline classifying 6+ Tabla stroke patterns into MIDI using **Mel-spectrograms**.
- Developed onset detection via **Librosa's beat tracking API** for time-aligned MIDI generation from raw audio.
- Deployed model as a **REST API via FastAPI** for real-time audio-to-MIDI inference in DAW workflows.

Emotion Recognition AI

TensorFlow, Computer Vision, Streamlit

Jul 2025 – Dec 2025

GitHub

- Trained a **CNN** with **transfer learning** on **35,000+ images** with precision/recall analysis across 7 emotion classes.
- Improved model robustness by **15%** via real-time **data augmentation** and Dropout regularization.
- Deployed as a **Streamlit** interface for real-time facial emotion classification from webcam or image input.

Achievements

GATE Data Science & AI (DA) and CSIT (Qualified)

Mar 2026

Proficiency in mathematics, probability, statistics, and Machine Learning through a national-level competitive exam.

AWS AI Practitioner Challenge

Apr 2026

Covers GenAI, LLMs, foundation models, prompt engineering, RAG, Amazon Bedrock, and responsible AI on AWS.

Hindustani Classical Sangeet Diploma

Jul 2019

Completed six years of Hindustani Classical Sangeet Diploma in Singing with multiple Distinctions.

Leadership & Extracurriculars

UDAAN - University Student Magazine

Graphic Designer

Feb 2025 – Present

Guru Ghasidas Vishwavidyalaya

- Created **30+ digital assets** using Adobe Suite and Canva to maintain brand consistency across channels.

EQUILIBRIO - Techfest

Cultural Coordinator

Mar 2025 – Apr 2025

Guru Ghasidas Vishwavidyalaya

- Directed a **50+ member team** to execute 10+ flagship events for a **2,000+ attendee** university techfest.